Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/22/2005
PI Name:	Oman, Charles M. Ph.D.		
Project Title:	Visual Orientation, Navigation, and Spatial Memory Countermeasures		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	coman@mit.edu	**Fax:**	FY 617-258-8111
PI Organization Type:	UNIVERSITY	**Phone:**	617-253-7508
Organization Name:	Massachusetts Institute of Technology		
PI Address 1:	Department of Aeronautics and Astronautics		
PI Address 2:	77 Massachusetts Avenue 37-219		
PI Web Page:			
City:	Cambridge	**State:**	MA
Zip Code:	02139-4301	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	03/01/2004	**End Date:**	02/28/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Harris, Laurence (York University) Liu, Andrew (Massachusetts Institute of Technology) Jenkin, Michael (York University) Aoki, Hirofumi (Massachusetts Institute of Technology)		
Grant/Contract No.:	NCC 9-58-NA00402		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The goal of this multi-institutional neurovestibular project is to develop four types of design, assessment, training, and procedural countermeasures: a) Evidence based spacecraft architecture and work area design standards. b) Methods for quantitative assessment of in flight and postflight oscillopsia. c) Preflight visual orientation training techniques to reduce disorientation and improve in flight emergency egress. d) Teleoperation procedure and training improvements based on crewmember spatial skills. Our goals are: 1) To quantify how environmental geometric “frame” and object “polarity” cues determine human visual orientation, to support engineering of spacecraft and work areas. 2) To develop reliable means for quantifying head-movement-contingent oscillopsia. 3 To determine whether preflight virtual reality techniques can improve astronaut 3D spatial memory and navigation abilities by reducing “direction vertigo”, and teaching ISS configuration and emergency egress routes. 4) To improve astronaut teleoperation performance by taking into account the mental object rotation and “perspective taking” abilities of individuals while training and during operations. Inflight spatial disorientation, spatial memory, navigation and teleoperation problems, and oscillopsia during re-entry and after landing have been identified as neurovestibular risks by Shuttle, Mir and ISS astronauts, NASA’s Critical Path Roadmap, the Neurovestibular Adaptation Team Strategic Plan, and a National Academy of Sciences committee report. NRA 03-OBPR-04 solicits research to determine what spacecraft architectures, interior visual cues, and preflight orientation training techniques will minimize inflight disorientation. The project utilizes the unique virtual reality research capabilities at York and MIT. Six sets of experiments and extensions are proposed: 1) Measuring the effect of environmental geometry (“frame”) cues using psychophysical techniques (York). 2) Assessing the influence of polarized objects on self-orientation perception using psychophysical judgments (York). 3) Assessing the extent and pattern of head-contingent oscillopsia and visual motion (York). 4) Effect of training module orientation on inflight direction vertigo (MIT). 5) Influence of relative body orientation in preflight visual orientation and egress training (MIT). 6) Correlation of spatial abilities with simulated space station remote manipulator training performance(MIT).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Results support the development of neurovestibular countermeasures for spatial orientation, navigation, and spatial memory difficulties among astronauts, and the design of future vehicles, including the CEV and Mars Transit Hab. Our results also pertain to human health on Earth, for example: origins and assessment of oscillopsia, and disorientation, spatial memory and navigation problems in vestibular, Alzheimer's and MS patients. Our results also pertain to the interior design of buildings to reduce disorientation by providing strong visual cues for orientation in both the vertical (to reduce falls in the elderly on stairs) and gravitational horizontal planes (e.g. the origins of “wrong door” phenomena in buildings and “geographic disorientation” in cities, and among sport orienteers, and in the design of visual cueing systems for civil and military flight simulators). To the extent that disorientation is reduced, motion sickness will also be alleviated.
Task Progress:	1) Previously, our principal research tools for studying frame and polarity effects were physical tumbling rooms. This year our 6 walled immersive IVY environment has become available. Our principal methods for assessment of the vertical included use of a tactile rod, or judging/adjusting the convexity of a shaded disc. This year we developed a new method, an Oriented Character Recognition Task (OCHART) involving the identification of a character whose identity is orientation defined (e.g. the letters “p” vs. “d”). Results (Vis. Sci. Soc. 2004) cross validate well with our other methods. For environmental manipulations in pitch, we evaluated having the subject set the orientation of a long virtual polygon (“virtual plank”). Comparing “up” judgments in roll using the IVY and the virtual plank vs in pitch using the physical room and the shaded disk method gave similar results. Using our previously published vector linear summation model (e.g. Jenkin Perception 2004), the plank method yielded a visual scene weighting coefficient of 37% of body+gravity in pitch, as compared to 45% using the shaded disk in roll in a physical environment. York is performing pilot experiments to study the effects of observer field-of-view, body orientation in pitch, number of polarized objects, and environmental aspect ratio. 2) Head movement contingent oscillopsia is commonly reported by astronauts during entry, landing, and for several days postflight, but methods for directly quantifying it have been lacking. Using VR techniques, we previously developed a technique (e.g. Jaekl JVR 2003) where the subject wearing a head mounted display adjust the gain of virtual scene motion during normal active movements until the scene appears stable. Normally subjects set visual gains somewhat greater than unity. York is extending this work to oscillopsia induced by Coriolis head movements in a rotating environment, and is constructing a display screen and head tracker mounted on a rotating chair. 3) Astronauts train in ground simulators, but physical constraints force the configuration of modules on the ground to be different from in flight. This year MIT conducted two experiments to see how survey knowledge of the interior of two virtual space station modules became impaired if the orientation of one of the modules was changed after training. Preliminary results were presented at the 2004 Barany meeting. 4) MIT constructed an immersive VR simulation of a space station interior, and is currently studying the influence of relative body orientation in a preflight visual orientation and egress training paradigm. The technique could 5) MIT initiated development of a robotic workstation simulator for Year 2 research, utilizing VRML ISS models as the virtual workspace, and software to present three or more workspace views and to compute robot inverse kinematics. We are evaluating alternative computer based tests of
Bibliography Type:	Description: (Last Updated: 12/18/2024)
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Dissertations and Theses	Benveniste D. "Cognitive Conflict in Learning Three-Dimensional Space Station Structures." Massachusetts Institute of Technology, September 2004. , Sep-2004
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Presentation	Liu, A. M. "A funny thing happened to me on the way to Mars... " Jul-2004
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